

Fixed effects and clustered inference

Slide route: Lecture10v2.tex: Fixed Effects Models through Final Remarks on Clustered SEs. Chapter 10.

Session 14 | Learn from changes within units

Outcomes: Compute a within transformation, interpret unit and time effects, state strict exogeneity, and explain clustering.

100-minute route

0–10	Retrieve what first differences remove.
10–30	Fixed-effects regression and within transformation.
30–45	Work the two-unit example.
45–55	Pause and contrast within/between variation.
55–75	Unit and time fixed effects; assumptions.
75–90	Serial correlation and clustered SEs.
90–100	Exit diagnosis and binary-outcome preview.

Slide-by-slide script

Slide 1 | Title

Guide Open with the question: what variation remains after removing a unit's fixed characteristics?

Slide 2 | Roadmap

Guide Preview within transformation, unit and time effects, strict exogeneity, serial correlation, and clustering.

Slide 3 | Last lesson

Guide Retrieve the first-difference equation and ask what it removes.

Slide 4 | Fixed effects regressions

Guide Define the unit effect α_i and explain why it can be correlated with X_{it} .

Slide 5 | Fixed effects regressions

Guide Calculate the unit means in the two-unit example: (2, 14) and (3, 26).

Slide 6 | Fixed effects regressions

Guide Demean X and Y within each unit and show the common within slope.

Slide 7 | Fixed effects regressions

Guide Explain that the within estimator uses changes relative to a unit's own average, not between-unit levels.

Slide 8 | Fixed effects regressions

Guide Ask what happens to a regressor constant within a unit. It is absorbed and cannot be estimated separately.

Slide 9 | Fixed effects regressions

Guide Contrast the within slope 2 with the between-unit slope 12 in the synthetic example.

Slide 10 | Fixed effects regressions

Guide Use a diagram to show why a large unit intercept difference can drive the between comparison.

Slide 11 | Fixed effects regressions

Guide Introduce time effects λ_t as common shocks shared across units.

Slide 12 | Fixed effects regressions

Guide Explain two-way fixed effects as unit effects plus time effects, with identification coming from remaining variation.

Slide 13 | Fixed effects regressions

Guide Ask whether fixed effects remove time-varying omitted variables. No.

Slide 14 | Raw panel data

Guide Label the unit and time axes before interpreting the cloud.

Slide 15 | Accounting for unit effects

Guide Show how subtracting unit means removes α_i .

Slide 16 | Accounting for time effects

Guide Show how time dummies absorb a common period shock.

Slide 17 | Both unit and time effects

Guide Explain that the two-way model can be overparameterized unless a reference convention is used.

Slide 18 | Both unit and time effects

Guide Ask which variables are absorbed by unit effects and which by time effects.

Slide 19 | Fixed-effects assumptions

Guide State strict exogeneity as a condition on errors across all periods, not only contemporaneously.

Slide 20 | Serial correlation

Guide Explain why differencing or repeated observations can make errors dependent over time.

Slide 21 | Clustered standard errors

Guide Use schools as clusters. Ask what observations may share an unobserved shock.

Slide 22 | Clustered standard errors

Guide Say that clustered SEs address inference under within-cluster dependence, not endogenous regressors.

Slide 23 | Final remarks

Guide Close with a checklist: source of variation, fixed effects, time effects, dependence, and number of independent clusters.

Worked example | Demeaning

Use with slides Lecture10v2 slides 38–49: Fixed Effects Regressions through the fixed-effects assumptions. Use the two-unit demeaning table with slides 38–43, then contrast it with unit/time effects on slides 44–49.

Use synthetic observations for two units over three periods:

$$X_A = (1, 2, 3), \quad Y_A = (12, 14, 16), \quad X_B = (2, 3, 4), \quad Y_B = (24, 26, 28).$$

The unit means are $(\bar{X}_A, \bar{Y}_A) = (2, 14)$ and $(\bar{X}_B, \bar{Y}_B) = (3, 26)$. After subtracting unit means, both units have

$$\tilde{X} = (-1, 0, 1), \quad \tilde{Y} = (-2, 0, 2).$$

Thus the within slope is

$$\hat{\beta}_{\text{FE}} = \frac{\sum_{it} \tilde{X}_{it} \tilde{Y}_{it}}{\sum_{it} \tilde{X}_{it}^2} = \frac{8}{4} = 2.$$

Contrast The line through the two unit means has slope $(26 - 14)/(3 - 2) = 12$. Between-unit differences and within-unit changes can tell very different stories. The large unit-level intercept difference explains this constructed contrast.

Assumptions and inference

Use with slides Lecture10v2 slides 45–54: accounting for unit effects, time effects, two-way fixed effects, strict exogeneity, serial correlation, clustered SEs, and final clustered-SE remarks. Use the school-clustering example during slides 50–54.

Start with

$$Y_{it} = \alpha_i + \lambda_t + \beta X_{it} + u_{it}.$$

Unit effects remove fixed differences; time effects absorb common time shocks. In a balanced panel, the two-way transformed variable is

$$\ddot{Y}_{it} = Y_{it} - \bar{Y}_i - \bar{Y}_t + \bar{Y}$$

and similarly for X . Use appropriate regression/projection machinery for an unbalanced panel rather than assuming this shortcut applies unchanged.

Strict exogeneity A familiar sufficient condition is $\mathbb{E}(u_{it} \mid X_{i1}, \dots, X_{iT}, \alpha_i) = 0$, with common time effects accounted for. Unobserved shocks that change later treatment choices can violate it. Fixed effects do not remove arbitrary reverse causality.

What cannot be identified A regressor constant within a unit is collinear with unit effects. A regressor identical across units within each period is absorbed by time effects. Ask which variation remains before interpreting a slope.

Clustering side example

There are 50 schools with 20 pupil observations each. Errors within a school may share a school-level shock. Treating 1,000 pupils as 1,000 independent pieces of information can overstate precision.

Say carefully Cluster-robust inference permits within-cluster dependence under its assumptions while relying on adequate independent cluster information. Fifty is not a universal adequacy threshold; few or unbalanced clusters can require special treatment.

Illustrative command For a genuine panel with identifiers `id` and time:

```
xtset id time
xtreg y x i.time, fe vce(cluster id)
```

Exit answers Fixed effects change the source of identifying variation. Clustered SEs address inference under dependence; they do not fix endogenous regressors. Adding time dummies does not absorb unit-specific changing confounders.

If short / preparation Reduce repeated diagrams while retaining the within calculation and clustering intuition. Read Chapter 11 and ask students why OLS might predict a probability above one.

Source: Lecture10v2.tex, fixed-effects and clustered-SE sections; synthetic examples.